
caitlinadams@ymail.com | London, UK | UK & Ireland citizenship

EXPERIENCE

CFC (Proactive Cyber), London, UK 2022 – Present

Proactive Cyber protects insurance clients against cyber risk, aiming to improve the book's loss ratio.

Data Scientist 2025 – Present

- Designed and established the business unit's first A/B tests, making experimentation the primary source of evidence for evaluating interventions in the £2m programme. Conducted power analyses, estimated a (conditional) treatment effect from experimental data, and extrapolated results to areas of the portfolio we couldn't directly test. Corroborated results with separate observational causal inference design.
- Developed novel, non-experimental survival analysis method for rare-event data to estimate the treatment effect of notifying customers of cyber risks on claims reduction, in the face of considerable selection bias and censored data. Used power analyses, minimum detectable effects, and negative outcome controls to communicate the result and residual uncertainty to senior stakeholders, ultimately resulting in the go-ahead for A/B testing.

Product Manager 2022 – Present

- Led the research, scoping and delivery of the division's core operational product from inception: a web application that ingests streams of risk data and automates client notifications about those risks. Grew the team from 1 to 4 engineers and tripled the volume of notifications delivered.
- Built the division's first medallion data layer on Databricks, feeding the senior-management dashboards used to make decisions about protecting 100k+ customers.
- Set up the division's data team and capability from scratch — hired and grew it from 1 to 3, staying hands-on with the analysis throughout (see above).

Accenture, London, UK 2021 – 2022

Senior Analyst

- Product Owner for a global retailer (~\$56bn annual revenue, 5,000+ stores), modernising the application that managed stock across their estate. Directed teams across the US, UK, and India.

Mott MacDonald, London, UK 2018 – 2021

Mott MacDonald is a global engineering, management, and development consultancy specialised in delivering large infrastructure and complex civil engineering projects.

Data Scientist

- Built and configured strategic transport models that informed multi-million-pound infrastructure decisions and government consultations, quantifying demand and traffic impacts across highways, rail, and cycling. Projects included HS2, Heathrow Expansion, TfL Rail Asset Management, TfL Cycleways, and EU Exit.
- Led the data-analytics workstream, and a four-person international team, producing the Average Annualised Daily Traffic figures underlying air-quality impact assessment. Automated the end-to-end process (Bash, VBA, Python, ArcGIS), cutting a ~3-day manual process to ~2 hours.

TECHNICAL SKILLS

Languages & tools: Python (pandas, scikit-learn, Spark), SQL, Bash, ArcGIS, Git, VBA

Statistics: A/B testing, power analysis, survival analysis, quasi-experimental research design, instrumental variables regression, negative outcome controls, selection bias and censoring

Platforms: Databricks, Microsoft Azure

EDUCATION

University of Birmingham, Birmingham, UK

MSci Mathematics — First Class Honours

2014 – 2018